Hospital Management System

Object-Oriented Programming Lab Mini Project

Group Number: 11

1. Abstract

This mini-project is a streamlined **Hospital Management System** developed using Java to demonstrate core Object-Oriented Programming (OOP) principles. The application provides a digital interface for hospital administrators to securely register and manage patient records. The system utilizes Java Swing for an interactive Graphical User Interface (GUI), allowing users to input patient details (Name, Age, and Illness) and view existing records dynamically.

2. Core Methodologies

To ensure data persistence without requiring an external database server, the system implements Java File Handling via object serialization. Patient data is automatically read from and written to a local `patients.dat` file, ensuring that records remain intact between application sessions. The architecture strictly adheres to OOP paradigms, utilizing classes and encapsulation to manage patient entities, providing a functional, reliable, and easily expandable administrative tool.

3. Implementation Details

The project satisfies the mandatory requirements set forth in the course syllabus:

- **Object-Oriented Programming:** Implements encapsulation by hiding data members in the `Patient` class and exposing them via getter and setter methods.
- **GUI:** Implements a user-friendly graphical interface utilizing the Java Swing framework (`JFrame`, `JTable`, `JTextField`).
- **File Handling:** Integrates `ObjectInputStream` and `ObjectOutputStream` to serialize objects directly to disk, ensuring data integrity without database connectivity.